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Aesthetic Sustainability and STEAM at Museums

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Abstract: *The aim of this article is to examine how aesthetic sustainability, as a theoretical concept and analytical framework, can elucidate and expand the science-technology-engineering-arts-mathematics (STEAM) approach to learning at museums, advancing it conceptually and practically. STEAM denotes an inter-disciplinary educational approach in which traditional boundaries between scientific and humanistic disciplines are deliberately blurred, allowing knowledge from each to intersect with and inform the other. The approach seeks to strengthen learners' capacity for critical thinking, creativity and problem-solving, while fostering awareness of the environmental, social and cultural implications of human action. It highlights how interdisciplinary and sensuous engagement with heritage materials may lead to transformative learning outcomes that connect scientific reasoning with aesthetic reflection. The article's overall argument is developed through analysis of four cases from the STEAM at Museums development project in Central Denmark Region (2023–25).*

Keywords: Aesthetic sustainability, STEAM approach to learning at museums, interdisciplinarity, transformative learning, museum objects and exhibitions.

English novelist and physicist Charles Percy Snow (1905–80) famously argued, in his book *The Two Cultures* (1959), that Western intellectual life is divided into two mutually incomprehensible cultures – the sciences and the humanities. He critiqued this separation and warned that the divide threatened the ability to address major global problems. He illustrated the gap by pointing out that literary intellectuals struggle to grasp scientific concepts, while scientists struggle to understand important literary works (Snow 1959, 4). The STEAM approach to learning unites the two paradigms, interconnecting scientific and humanistic disciplines. In many ways, this interdisciplinary approach is fundamentally about sustainability. It seeks to strengthen students' critical thinking and creative problem-solving skills and encourages them to evaluate the environmental, social and cultural impact of their actions and decisions. It forms the basis of process-oriented learning activities focused on real-world issues

that often directly or indirectly address sustainability challenges.

The question this article aims to examine is how aesthetic sustainability as a theoretical concept and an analytical framework can serve to elucidate the science-technology-engineering-arts-mathematics (STEAM) approach to learning at museums, advancing it both as an idea and in practice.

This article contributes to the body of research dealing with aesthetic sustainability (Harper 2018, Lehtinen 2021, Grøn and Christensen 2022, Grøn and Jantzen 2022,) as part of the overarching topic of sustainability at museums (Stylianou-Lambert et al. 2014, Kristinsdóttir 2017, Saraniero and Kelton 2019, Evans and Achiam 2021, Lehtinen 2021). It does so by suggesting how a selection of theoretical perspectives pertaining to aesthetic sustainability can be used to elucidate the STEAM approach to learning in a museum setting.

In the STEAM at Museums development project in Central Denmark Region (2023–25)[1], the STEAM approach served as the premise for the coursework that museum educators and upper secondary school teachers developed for the students to work on at the museum. Four of these coursework activities, as they are presented in the final project report (Vølund and Porse 2025), have been selected as case material for this study.

What is aesthetic sustainability?

The understanding of aesthetic sustainability in this article is mainly based on how art historian and philosopher Sanna Lehtinen and especially design historian Kristine H. Harper interpret the concept. According to Lehtinen, aesthetic sustainability directs attention toward the more profound dimensions of aesthetic appreciation, which integrate aesthetic, ethical and cognitive values (Lehtinen 2021, 255). Short-lived fads and fashion do not belong to these more profound dimensions nor adhere to such values. Viewed from an aesthetic perspective, the long-lasting and genuinely impactful object stands at odds with passing trends (Harper 2018, 87).

Research deals with the relationship between aesthetics and sustainability from many different perspectives, e.g. the aesthetics of nature, environmental aesthetics, everyday aesthetics, or design aesthetics (Grøn and Jantzen 2022, 10). The perspective in this article adheres mainly to the latter, as design aesthetics is Harper's research domain. Although Harper's discussion of aesthetic sustainability is centred around the life cycles of objects produced for everyday use, e.g. furnishings, tableware, clothing and arts and crafts, it is also relevant in relation to museum objects. Most museum objects have had a "life" outside the museum, many as everyday objects. As museum objects they are different from everyday objects because they are no longer in use in their original capacities – a circumstance that extends their life cycles indefinitely. In the museum, they are on display; their main function now is to be containers of knowledge and – as the object of museumgoers' gaze – to contribute to the museum's curated narrative. The museum object thus becomes the object of aesthetic appreciation and is the embodiment of temporal durability.

Sustainability is essentially about temporal durability and endurance (Lehtinen 2021, 261). This understanding is a core element of three established, interconnected categories of sustainability: environmental, economic and social. It also underpins a fourth and more

recently established category, namely cultural sustainability, defined as the recognition, safeguarding and showcasing of tangible and intangible heritage of diverse social groups, communities and nations (Stylianou-Lambert et al. 2014, 568). The preservation of cultural heritage and the promotion of cultural vitality and viability are key factors of cultural sustainability (Soini and Birkeland 2014, 8). Museums, serving as custodians of many types of heritage, can foster sustainable development by providing education on sustainability, e.g. preserving, interpreting and activating the past to enlighten the future.

Aesthetic sustainability, as a fifth and more recent category, also has "temporal endurance and durability" at its core. Durability refers not only to durable materials and sustainable design practices, but also to "emotional durability" in the sense of the emotional, sensory and long-lasting connectedness between subject and object. "Aesthetically sustainable objects continuously add nourishment to human life" (Harper 2018, abstract), which is why humans go to such lengths to care for them and preserve them and – potentially – to produce objects of similar durability for coming generations.

Harper adopts three concepts of time that are closely linked to her understanding of aesthetic sustainability. These three concepts are "time of becoming", "time of existence" and "time of being"[2] (Harper 2018, 88ff), and they are particularly relevant for the analysis of STEAM learning at museums.

"Time of becoming" refers to the time that elapses in the process of growing or maturing from material to finished product. Often, a product's value is enhanced when its creation involves careful forethought and craftmanship. Furthermore, the subject's sensory engagement with the object can foster a deeper, more enduring involvement with it, also increasing its value. Indeed, the "time of becoming" can extend beyond the actual process of creation. Through its reception and use, an object's value can continue to increase long after it has achieved its physical form. Such an object, in Harper's words, "is highly aesthetically sustainable" (Harper 2018, 88).

"Time of existence" denotes the period through which an object has endured since its creation, manifested through the "marks of time" it bears; over time, it may age, fade or deteriorate. According to Harper, marks of time often add to an object's value because they are proof of its ongoing functionality and performance. Like the beauty of a careworn face, the passage of time enriches an object's depth and character, highlighting rather than eroding its intrinsic qualities (Harper 2018, 92). Again, the link to aesthetic sustainability is clear.

"Time of being" reflects the dynamic engagement between subject and object, which happens when and while the subject is taking it in or perceiving it. The focus of "time of being" is the lived aesthetic experience itself. Harper argues that aesthetic experience involves self-experience: by fully engaging with and attending to the world through their senses, people enrich their phenomenological understanding and enhance their aesthetic intelligence. This sensuous awareness can be further developed through deliberate sensory experiences. The experience of the tactile and visual characteristics of an object can inspire an aesthetic connection between subject and object. This connection or attachment can arise from a prolonged engagement with the object (Harper 2018, 93–94). It seems clear that the temporalities of these three concepts can be applied to museum objects.

Museum objects are cared for meticulously. They often inspire awe and compel deep interest because of their age, their beauty, their cultural or political significance, the skill with which they were manufactured, or many other factors (Thorhauge 2021, 277–78). Many of the objects in a museum represent fine artisanship or artistic skill; objects of poorer quality perish and never make it into a collection. Museum objects contain tangible and intangible aesthetic qualities worthy of emulation and further development (Stylianou-Lambert et al. 2014, 566). In other words, museum objects contain the temporalities of Harper's three concepts.

Lehtinen maintains that aesthetic sustainability offers an understanding of why certain objects and environments are valued more than others and how this appreciation evolves over time (Lehtinen 2021, 255). According to Lehtinen, Harper's design theory provides particularly valuable insights into aesthetic sustainability. Harper points out that objects as vessels of time carrying physical, material or tangible narratives gain emotional and sensuous significance for the recipient or user (Harper 2018, 85). Museum objects are brimming with such narratives. They have undergone a process of musealisation and are intrinsically imbued with greater value than other objects. Digging through these museal layers of meaning and unravelling an object's value and narratives are activities well-suited to education for sustainable development (Evans and Achiam 2021, 1192), including STEAM.

I now very briefly present the STEAM at Museums development project, followed by an exploration of aesthetic sustainability in relation to museum objects in the first three STEAM cases.

STEAM at museums: museum objects and aesthetic sustainability

The development project was a two-year initiative (August 2023 – June 2025) that explored how museums can serve as learning resources and environments for STEAM education at the upper secondary school level.

Seven museums and nine schools across the Central Denmark Region formed nine partnerships facilitated by the interface association.[3] The project had three main aims. First, it sought to strengthen students' science and arts competencies through interdisciplinary briefs anchored in authentic museum challenges. Second, it created process- and practice-oriented learning experiences in which students collaborated across subjects – such as chemistry and the visual arts – to apply disciplinary insights and propose concrete, sustainable solutions to the museum-based briefs. Third, it supported the professional development of teachers and museum educators, who jointly tested and refined new pedagogical tools for STEAM teaching and learning in museum contexts. An important premise of the project was that museums offer a useful setting for interdisciplinary learning, providing access not only to artistic, cultural, historical and scientific artifacts, but also to knowledge about exhibition design, conservation and dissemination.

I now turn my attention to three partnership projects, each of which approached museum objects from a distinct angle: as emotional and therapeutic triggers (Museum Ovartaci), as sensory gateways combining chemistry and literature (Skovgaard Museum), and as inspirational frameworks for experimental reinterpretation (Holstebro Art Museum). In other words, each case illustrates a different way of bridging the gap between disciplines – and be-

tween students and objects. I apply my understanding of aesthetic sustainability in relation to each, to disclose how the concept clarifies the STEAM approach.

Art as a gateway to understanding neuropsychological disorders

In this project, carried out by Museum Ovartaci and Aarhus Katedralskole, advanced-level biotechnology students participated in Lundbeck's annual Drughunter competition by producing posters that addressed a series of neuropsychological disorders.[4] A distinctive element of this project was the requirement to engage with art by Ovartaci (1894–1985), an artist with a history of psychiatric disorder, whose work forms the core of Museum Ovartaci's collection. As part of the project, students visited the museum and, in small groups, engaged with one of three artworks through paraphrasing and therapeutic art exercises. These arts-based methods offered alternative humanistic perspectives on mental health, complementing and illuminating the scientific dimension of the project. The project culminated in collage and poster presentations at the museum, attended by museum staff, the students' teacher and the guests who happened to be there at the time (Vølund and Porse 2025, 28–29) (fig. 1).

During the project, the students learned about and connected with topics and spaces that had previously been unfamiliar to them: Ovartaci's art and mental illness and the museum, respectively. Harper points out that aesthetically sustainable value can be ascribed to an object, and that this can serve as the basis for forming a bond between people and things (Harper 2018, 116). A bond such as this might describe the one forged between the students and the Ovartaci art, arising out of their interpretive study of the work, and then combining their insights with what they learned about the artist's mental illness specifically and neuropsychological disorders in general – and how to treat them – in their biotechnology lessons. Ovartaci's artwork, as well as his mental disorder, were windows through which the students gained a more nuanced understanding of the science underlying treatments for psychiatric disorders.

Lehtinen claims that aesthetic sustainability emphasises deeper layers of aesthetic appreciation, integrating aesthetic, ethical, and cognitive values (Lehtinen 2021, 255). This offers a perspective on the Ovartaci students' process. Aesthetic and cognitive activities were embedded in their coursework, in that they used their science and arts subjects to draw close to Ovartaci's artwork, understanding how it was an expression of his mental state. Harper's "meaningful bond between subject and object" (Harper 2018, 116) describes how the students, in the course of and as a result of their work processes, might also have formed a bond – an emotional attachment – with the artist, as another type of "object", and become ethically engaged in how his life as a patient for five decades in a mental health facility played out.

Considering the students' work and learning through Harper's and Lehtinen's thoughts, it is possible to imagine that the students learned something about how and why, on the one hand, mental illness requires careful medical treatment and, on the other, how it could and can be the source of great creative energy. Bonding with the artwork and the artist during

Fig 1. A student in the process of making a paraphrase of a work of art by Ovartaci. Photo courtesy of Museum Ovartaci.


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their prolonged study would contribute to this learning. Harper points out that involving the subject in the process of creating a product – in this case a collage that required physical interaction with the materiality of the product – helps create a bond that imbues the product with special value (Harper 2018, 88, 116). This is what Harper calls "time of becoming", and in relation to the Ovartaci students' work, it sheds light on how creating a personal collage could – vicariously – deepen their understanding and appreciation of Ovartaci's own creative processes as are expressed in his works of art.

The collage activity can furthermore be characterised as Harper's "time of being", as it represents a deliberate sensuous experience in the students' dynamic, ongoing engagement with and perception of Ovartaci's artworks. The students' aesthetic experience involved, quite literally, a significant degree of self-experience. This also ties in with Lehtinen's suggestion that profound aesthetic appreciation is an integral part of aesthetic sustainability (Lehtinen 2021, 255).

Golden Age painting reinterpreted through scent and sound

This project invited students from Viborg Gymnasium, as part of their chemistry and Danish lessons, to revitalise Skovgaard Museum's Golden Age period collection by adding new layers of interpretation based on sensory experiences. The students worked in groups, and each group received a painting selected by the museum. As their output in chemistry, each group identified a motif or element in their painting that they could associate with a particular scent and then constructed a 3D molecular model of that scent. Their output in Danish was to produce a written as well as an audio-formatted interpretation of the painting. These outputs were then exhibited at the museum at the project's closing event. According to the description of this project in the STEAM at Museums evaluation report, there were indications that student engagement deepened because their task was to contribute to the museum's interpretive landscape and because their contributions derived from readings based on bodily senses that transitioned into novel interpretations of the works of art (Vølund and Porse 2025, 30–31) (fig. 2).

In contrast to the Ovartaci project students' process, where students explored neuropsychological disorders from a humanistic angle, the students' starting point at Skovgaard was a work of art that they were required to explore and interpret in a new way, using not only Danish but also, unconventionally, chemistry as their disciplinary perspectives. Working with the visual arts is part of the syllabus for Danish, so applying this subject in their interpretive work with the paintings would not have been foreign to them. Golden Age paintings, however, are not typically incorporated in lesson plans for chemistry. It is the latter that is interesting to reflect on in light of Harper's thoughts on "the pleasure of the unfamiliar" (Harper 2018, 33ff). According to Harper, some objects cannot be immediately understood; they disrupt our instinct to categorise and organise, forcing us to engage fully with the present moment and expanding our consciousness. Although such challenging experiences can be uncomfortable, this discomfort carries a distinct pleasure, as humans have a fundamental need to be pushed and have their assumptions challenged. This balance of challenge and engagement gives rise to the pleasure of the unfamiliar, making challenging objects aesthetically sustainable (Harper 2018).



Fig. 2. 3D molecular models of the scents identified in Golden Age paintings, exhibited at the museum. Photo courtesy of Skovgaard Museet.

Although the objects Harper is discussing are practical objects designed for use, her line of reasoning regarding the challenging mental processes that take place when dealing with disruptive objects sheds light on the mental processes of students whose task was to engage with and interpret a work of art from the perspective of chemistry. Identifying a motif in a Golden Age painting that could be translated into a scent and then further translated into the language of chemistry – a 3D molecular model representing that scent – would have obliged the students to "think outside the box" and apply their chemistry knowledge unconventionally. Learning in such circumstances is in fact enhanced, because students are compelled to adapt their learning strategies in innovative ways or engage a broader repertoire of strategies than classroom instruction generally supports (Thorhauge 2014, 224). Harper's claim that challenging objects are aesthetically sustainable could therefore perhaps be linked to their educational value. Expressed differently, it seems clear that Harper's understanding of aesthetic sustainability in this context is closely connected to impactful learning. This is also indicated in Harper's point that a "sublime aesthetic experience" (Harper 2018, 34) is often the result of challenging and provoking encounters with things in our surroundings; and although

these encounters may be fleeting in character, they leave enduring traces in the experiencing subject's mind (Harper 2018, 34–35).

Learning of this kind is "transformative", and it entails guiding students to reflect critically on assumptions underlying implicit values and beliefs (Taylor and Taylor 2019, 128–29). One way of doing this is to strengthen their "cultural-self knowing" (Taylor and Taylor 2019, 128–29). This can be done by creating learning contexts that allow students to understand how humans' culturally shaped selves – and the largely hidden assumptions underlying their shared values, beliefs and ideals – form their cultural identities and influence their habitual ways of being, understanding and interacting with the social and natural environments (Taylor and Taylor 2019, 128–29). Interpreting a Golden Age painting by means of chemistry (and Danish) is, of course, only a very small example of this. However, applying an interdisciplinary approach in this way allowed the students to recognise that the natural world represented in painting was the same as the one represented in their text, audio and molecular models, even though each medium interpreted it in a very different way (Vølund and Porse 2025, 31). Alternative and perhaps hitherto unimagined interpretations of the natural world of the painting thus became apparent to the students. According to Harper's thinking, their assumptions were challenged, making nuanced learning possible.

A chamber of wonder as an aesthetic laboratory

Over 250 visual arts students from Holstebro Gymnasium participated in this large-scale project centred around the installation *Chamber of Wonder*[5] (2001) by Danish artist John Olsen (1938–2019). The installation, which is on permanent display at Holstebro Art Museum, is a series of cabinets displaying specimens, everyday objects and his own artwork, using systematic display and assemblage techniques to explore decay, nature's destructiveness and the aesthetic beauty of decomposition. Students began their work at the museum with a guided tour (a local artist was their guide) and a collection activity in a forest close to the museum, which was performed as an homage to Olsen's own fieldwork methods. They gathered many types of materials, i.e. bark, branches and twigs, pieces of processed wood, cardboard, metal, glass, and litter. Back at school, the students combined these materials with scientific experiments, e.g. redox reactions on the pieces of metal, carried out under the guidance of a teacher trained in both chemistry and visual arts. In this project, the two subjects truly merged, as the students explored scientific phenomena using aesthetic and artistic methods, and vice versa, once again imitating Olsen.

They were then invited to curate an exhibition of their objects, organising them in a collective installation – a gesamtkunstwerk – which was displayed at the museum as their own interpretation and paraphrase of Olsen's work of art. This was a way to revitalise Olsen's installation. The entire learning experience was organised as a series of process-oriented and exploratory activities, whereby the students re-visited the artwork and their own work many times, in contrast to more traditional result-oriented assignments (Vølund and Porse 2025, 34–35) (fig. 3).

Harper maintains that aesthetic sustainability involves fostering enduring emotional and sensory bonds between people and objects and places, respectively, through meaningful aes-



Fig. 3 The students' installation "Wonder" ["Underværk"]. Photo by Sara Holm Strandby, courtesy of Holstebro Kunstmuseum.

thetic engagement – as an expanded dialogue (Harper 2018, 94). The students' collecting activity in the forest, which served both to connect them to Olsen's artwork and to open their eyes to their local environment, can be perceived as meaningful aesthetic engagement. The act of collecting outdoors in a local forest would have involved physical senses (e.g. looking, touching) and required physical activity (e.g. walking, bending over, picking up) as well as social interaction (e.g. talking among classmates, showing finds to one another, consulting with their guide). As a school-related activity, all of this would have been a novel experience for the students in these surroundings, and the effect of the novelty would likely have been a sharpening of their senses, their attention and their powers of observation (Thorhauge 2014, 222). Harper calls an experience like this "taking in the world" (Harper 2018, 94), which is also an element of "time of being" – in this case, the time the students spent with the objects (the finds) in their surroundings (the forest). According to Harper, the "time of being" can be extended by immersing ourselves in the process of apprehending the surrounding world and its objects.

By being fully alive to our environment – sensing, interacting with and becoming absorbed in the things in our surroundings: taking them in, lingering with them and savouring them – we nourish our bodies with phenomenological insight. The more sensuously present we are, the more attuned and aesthetically perceptive we become (Harper 2018).

"Time of being" in this project was further extended and strengthened during the transformation of the students' collected objects into "art" by means of diverse processes of experimentation, refinement and adaption; the students' repeated visits to the museum to immerse themselves in Olsen's installation; and finally, the curation of their installation and exhibition. The actual creation of their own "works of art", and then later collaborating to curate and exhibit them, represents Harper's "time of becoming", whereby an emotional bond between students and both their art and their exhibition would have been formed and strengthened. Harper's concepts "time of becoming" and "time of being" suggest that the Holstebro students' immersive, sensuous engagement, hands-on experimentation and exhibition curation – coupled with their reflections about Olsen's *Chamber of Wonder* – deepened their appreciation of the artist's process as well as of their own aesthetic experience. This supports Lehtinen's view that intense aesthetic engagement underpins aesthetic sustainability (Lehtinen 2021, 255).

Harper's third concept, "time of existence", is also relevant at this point. Olsen spent many years collecting an enormous array of objects for his work of art. A large number of these objects were worn, skeletal, decayed, desiccated, rusty, chemically preserved, had been discarded and were no longer of any practical use, or bore witness to the passage of time in other ways. In the mundane world, many of the objects would have been categorised as worthless. But in their transformation into art, the objects' signs of existence became signs that they had been "loved" (Harper 2018, 91) by the artist, who not only collected them, thereby saving them from non-existence, but also recognised their inherent artistic potential, combining and utilising their colours, dimensions, textures, stories and histories – in short, their materialities and immaterialities – to produce a work of art. In the art museum, they took on aesthetic and museal value and insisted on being viewed and understood from completely new perspectives. These objects had once again gained viability and vitality. Through their imitation of the artist's methods (collecting objects, experimenting and in some cases chemically treating and modifying them, curating and displaying them), the students acquired an insight into the aesthetic transformation that took place in relation to their own objects. According to the evaluation report, the students understood that they were learning in a new way because they were challenging themselves and creating their own processes, modelled on Olsen's artistic methods (Vølund and Porse 2025, 35). Like Olsen, they focused on what Harper calls "the beauty of changeability" (Harper 2018, 91), adding aesthetic value and, not least, durability to the objects they had found. Importantly, the students' paraphrase of Olsen's installation shed new light on it and added yet another chapter to its "time of existence".

Lighting in museums, or museum exhibitions and aesthetic sustainability

In the fourth project, carried out by the Museum of Eastern Jutland (MEJ) and Randers

Statsskole, focus was shifted away from museum objects and instead directed at the investigation and interpretation of the museum's exhibition design. The museum was not merely a setting for learning but became the object of inquiry in its own right, inviting students to develop an aesthetic and reflective relationship with the institution and its exhibition design practice.

The project involved students from a biology class and from a design and architecture class. The biology students investigated how the dim lighting used to protect sensitive objects could lead to museum fatigue, and they explored how these adverse physiological and mental effects could be mitigated. The design students developed proposals for new lighting designs for the permanent exhibitions, focusing on both aesthetic design and functionality. Their challenge was to create pendant lamps made of recycled materials and tailored for different areas of the exhibition, keeping in mind conservation constraints and visitor wellbeing. The design of the lamps was furthermore to underscore the chronological timeline of the exhibition.

The students' work was exhibited at the museum at the end of the project. Design students showcased their lamps at the museum's entrance, accompanied by screens explaining the project's phases. Biology students contributed a "Light Safari", with floor-mounted texts explaining how museum lighting, light and sleep hormones impact mood and wellbeing (Vølund and Porse 2025, 32–33) (fig. 4).

The sustainable bond that can be created between a subject and an object through aesthetic engagement can be expanded to include enduring emotional and sensory connections between people and places (Harper 2018, 94). Lehtinen points out that aesthetic sustainability offers an understanding of why some environments are assigned greater value over time (Lehtinen 2021, 255). Encouraging students to design considered solutions for the museum's exhibition space, the coursework for the project demonstrated how such connections could be meaningfully stimulated. Here it is again relevant to mention Harper's concept "time of becoming", which was manifested in the students' creation of lamps and a "Light Safari", respectively. The lamps and the Light Safari were the product of the students' interaction

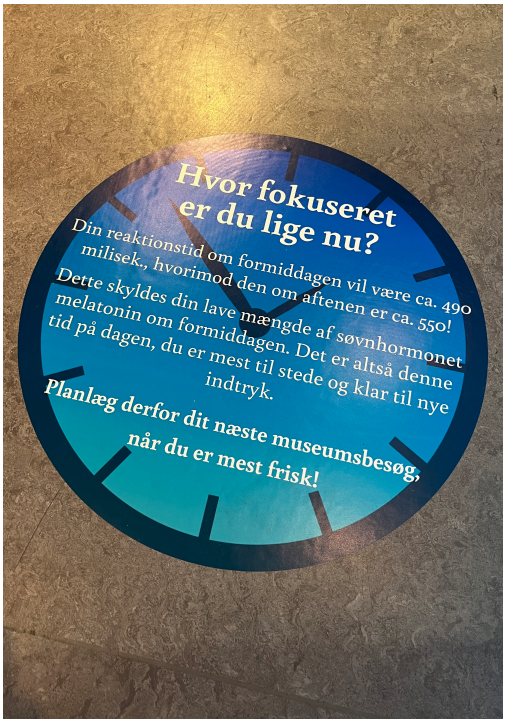


Fig. 4 "How focused are you right now?" This is one of the large round stickers of the Light Safari. It explains that the sleep hormone melatonin is lowest during late morning, meaning that a person is at her most alert at that time of day. It then encourages guests to visit the museum when they feel most alert. Photo by Line Højgaard Porse.

with the museum space or, more specifically, the *lighting* used in the exhibitions, which was the actual object of their study and interpretation. Another important, though less available, object of study was the museum guest, or, more precisely, the eyes and the visual experience of the museum guest. This was of special interest to the museum and was part of the brief. Because the MEJ project brief was authentic and the students' solutions were given an authentic afterlife at the museum – as in the other projects – the students would have been strongly motivated to do the work (Thorhauge 2014, 223), becoming even more profoundly invested in the museum and its space in the process. According to the evaluation report, the students were very proud of the fact that their lamps and floor stickers were displayed at the museum (Vølund and Porse 2025, 33).

An overarching theme of the brief for the students was sustainability. Both groups of students learned about why displayed museum objects must be scrupulously protected against light-induced deterioration. The design students then designed and produced sustainable lamps for several historical exhibitions. These lamps not only took light-sensitive museum objects into consideration but were also largely comprised of recycled materials and symbolised in their design the exhibitions' historical periods. Visitors could view the lamps and read about the students' process, thus learning about museum practice regarding lighting in historical exhibitions. The biology students learned about how and why the museum's necessarily dim lighting impacts museum guests' eyes and, consequently, their (visual) museum experiences. The students' solution – the Light Safari – was meant to provide insight about exhibition lighting to the guests and thereby, possibly, improve their experience. It could perhaps even be understood as a way of improving guests' "time of being" at the museum.

Harper's concept "time of being" sheds light on the dynamic interaction between subject and object during perception, focusing on the lived aesthetic experience, where fully engaging the senses enriches self-understanding and enhances aesthetic intelligence. This angle clarifies both the students' – and, ideally, the guests' – lived aesthetic experience of the space of the museum.

Concluding thoughts about aesthetic sustainability and STEAM

An object in a museum can be seen as being "long-lasting and genuinely impactful" (Harper 2018, 87). Lying in its display case, standing on its pedestal or hanging on the wall, the object always marks the end of an ever-lengthening past, and it is *right here* that it meets the museum guest. Then, for a space of time, the object and the guest accompany each other. How long this space of time lasts depends on the impact the object has on the guest. In theory, the space of time can be sustained until the guest departs this earth. This stretch of time corresponds to the "time of being", one of the three concepts of time related to Harper's understanding of aesthetic sustainability. The concept "time of being" relates well to the principles of STEAM. Museum objects provide insights into many domains of knowledge, which makes them ideal for immersive study in a STEAM context. Students' prolonged involvement with objects gives them time to apply their various cognitive skills to the tasks at hand, and, just as importantly, to experience the object or exhibition with their emotions and senses, fostering an emotional and intellectual connection with it (Harper 2018, abstract).

Such motivating, multifaceted learning is impactful, meaning that the "time of being" could stretch past the end of students' STEAM projects and quite possibly contribute to a more nuanced understanding of the importance of applying both scientific and humanistic approaches when working to solve complex problems.

Museum objects are containers of what Harper calls "thing-magic", which can be understood as the power objects wield because of the stories buried within them (Harper 2018, 85–86). An object's "thing-magic" is closely related to its "time of existence", which also resonates with STEAM learning. By applying knowledge from their science and humanities subjects during their engagement with museum objects, students can gain a thoughtful understanding of how objects represent histories and stories other than their own, thus giving them a sense of their own place in the larger order of things. Their aesthetic appreciation of the objects becomes more discerning, which, according to Lehtinen, is an important element of aesthetically sustainable thinking (Lehtinen 2021, 255). Students learn that museum objects – collected and preserved as cultural heritage for posterity – encapsulate a temporality that exceeds their own.

Also relevant to STEAM is Harper's assertion that engaging the subject in the act of creating a product can generate a lasting relationship between object and subject, as it encourages her to explore and interact with the object through touch and sight (Harper 2018, 88). This corresponds to Harper's concept "time of becoming", and it sheds light on how students doing STEAM projects are required to work. The STEAM approach emphasises the importance of strengthening students' ability to apply interdisciplinary knowledge to create innovative solutions to complex, real-world problems. Although coursework that tasks students with creating a product that solves an authentic problem at a museum is only a small-scale version of this, students doing such coursework are conducting themselves in the way the STEAM approach requires.

Echoing C. P. Snow's warning about how the divide between the sciences and the humanities threatens humans' ability to address major global problems, an ambition of the STEAM interdisciplinary approach is to stimulate the understanding that sustainability is a complex issue that cannot be solved by technological advances alone. The climate crisis is not merely an isolated environmental phenomenon, but also a social and cultural crisis, so effective climate policy and sustainable practice must change the cultural patterns and socio-economic logics that caused it (Grøn and Jantzen 2022, 8). In other words, an ambition is to foster an awareness of the need for what Ulrich Beck calls "cosmopolitan solidarity" (Beck 2010, 255), which denotes the commitment to uphold the global common good, even when doing so conflicts with immediate personal interests. Implied is an ethical imperative, the urgency of which is also emphasised in Sanna Lehtinen's reflections on the crucial importance of making sustainably responsible aesthetic choices (Lehtinen 2021, 256). Recognising the extent of both individually distinct and collectively held aesthetic values is significant for areas of sustainable development that pertain directly to human life and experience. This again encompasses a wide spectrum of human existence, including everyday living environments and patterns of consumption (Lehtinen 202, 257).

While the conceptual compatibility between aesthetic sustainability and the STEAM framework appears strong, its practical implementation may reveal certain tensions. The ongoing inflexible emphasis on demonstrable outcomes

in formal education could stifle the ethical and emotional dimensions that Harper's temporal concepts seek to strengthen. Likewise, implementing activities that depend on prolonged study may not easily align with the structured timelines and performance metrics characteristic of formal education. These tensions, however, need not be regarded as insurmountable barriers; rather, they point to the necessity of refined pedagogical design and critical reflection on how aesthetic experience, interdisciplinary inquiry and process-oriented coursework focused on authentic issues can be meaningfully put into practice in museum education programmes.

Through interdisciplinary and hands-on engagement with museum objects, students seem to develop a deeper understanding of the expanse and effects of time, of the communicative and temporal nature of objects and exhibitions, and of aesthetic choices, as well as reflecting critically on implicit assumptions – all of which is addressed and illuminated by the concept of aesthetic sustainability. This type of new insight represents transformative learning (Taylor and Taylor 2019, 126ff). Transformative learning seems to be a key component of the thinking behind aesthetic sustainability as well as of the principles of STEAM learning. Museums might re-think their general interpretative programmes, with aesthetic sustainability as their ideological point of departure (Saraniero and Kelton 2019, 545–46) and include STEAM projects as one of these programmes. In so doing, museums could make a significant contribution to sustainability education. Further research could explore how aesthetic sustainability might be operationalised in museum education programmes and assess its long-term impact on learners' capacity for interdisciplinary and ethically informed thinking.

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Notes

1. For a short description of the STEAM project, please see https://intrface.dk/om-intrface/steam/. To read more about the project, please see the article "Wicked problems i billedkunst og design – kan de løses med STEAM-undervisning og inddragelse af museer?" ["Wicked Problems in Art and Design – Can they be solved by teaching STEAM at museums?"] in *Magasinet TEGN*, 2025: 60–65. For a more comprehensive description of the project, please see the publication *STEAM på museer* [*STEAM at Museums*], edited by Stine Vølund and Line Højgaard Porse. Viborg Gymnasium 2025.
2. These three concepts of time were originally articulated by artist Willy Ørskov (1920–90). Harper appropriates the concepts because, as she says, "they are highly relevant in regard to exploring aesthetic sustainability" and "have served to structure the discussion about how to work with charging an object with time" (Harper 2018, 87). https://intrface.dk/
3. Drughunters is an educational program in biology, chemistry, biotechnology and mathematics, where students work as researchers and develop new ideas for treating brain diseases (https://www.lundbeck.com/drughunters).
4. The artist John Olsen coined a word, namely "Undrekammer", for the title of his installation. Translated literally, this means "a chamber to wonder in", or, as suggested here, "chamber of wonder".

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